Divisibility by Pairing, Finite Differences, and Harmonic Reflection

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Three problems in this note appear unrelated: sums of odd powers, an exponential congruence modulo 512, and an alternating rational sum whose numerator is divisible by 1979. Their solutions become shorter once the expressions are organized by symmetry, telescoping and modular inverses. The number-theoretic conventions follow MIT 18.781 *Theory of Numbers*; for finite differences, see also Graham, Knuth and Patashnik, *Concrete Mathematics*.[1]

§1.1 Odd Powers and Triangular Numbers

Theorem 1.1.1

For every integer $k \geq 1$ and odd positive integer r,

$$\boxed{\frac{k(k+1)}{2} \mid \sum_{j=1}^{k} j^r.}$$

Proof. Write $S = \sum_{j=1}^{k} j^r$. Since r is odd,

$$(m-j)^r \equiv -j^r \pmod{m}.$$

If k is even, pairing j with $k+1-j$ has no fixed point, giving $k+1 \mid S$. Modulo k, the terms $1, \ldots, k-1$ are paired by $j \mapsto k-j$; the only possible fixed point is $k/2$. Hence $k/2 \mid S$. As $\gcd(k/2, k+1) = 1$,

$$(k/2)(k+1) \mid S.$$

If k is odd, instead pair $1, \ldots, k-1$ modulo k, with no fixed point, and note $k^r \equiv 0 \pmod{k}$. Thus $k \mid S$. Pair the terms $1, \ldots, k$ under $j \mapsto k+1-j$; the unique fixed point $(k+1)/2$ contributes a multiple of $(k+1)/2$. So $(k+1)/2 \mid S$. Since $\gcd(k, (k+1)/2) = 1$, their product divides S. □

The parity cases matter. For $k = 3, r = 1$, the sum is 6, divisible by $k(k+1)/2 = 6$, but not by $k+1 = 4$. A proof that claims all terms can simply be paired modulo $k+1$ overlooks the central fixed point.

§1.2 The discrete fundamental theorem of calculus

For an integer-valued sequence f, define the finite difference $\Delta f(n) = f(n+1) - f(n)$. Telescoping yields

$$f(n) = f(1) + \sum_{j=1}^{n-1} \Delta f(j).$$

[1] https://ocw.mit.edu/courses/18-781-theory-of-numbers-spring-2012/pages/lecture-notes/

Thus if $m \mid f(1)$ and $m \mid \Delta f(j)$ for every $j \geq 1$, then $m \mid f(n)$ for every $n \geq 1$. This is induction expressed in a form that separates the initial value from the repeated increment.

The operator also reduces polynomial degree:

$$\Delta n^2 = 2n + 1, \qquad \Delta \binom{n}{k} = \binom{n}{k-1}.$$

The binomial basis is especially convenient because taking a difference simply lowers the index. For a polynomial P of degree d, $\Delta^{d+1} P = 0$.

§1.3 A congruence modulo 512 in two steps

We wish to prove

$$\boxed{512 \mid 3^{2n} - 32n^2 + 24n - 1 \qquad (n \geq 1).}$$

Let $F(n) = 9^n - 32n^2 + 24n - 1$. The base case is $F(1) = 9 - 32 + 24 - 1 = 0$. Its finite difference simplifies to

$$\begin{aligned} \Delta F(n) &= 9^{n+1} - 9^n - 32((n+1)^2 - n^2) + 24 \\ &= 8(9^n - 8n - 1). \end{aligned}$$

Set $H(n) = 9^n - 8n - 1$, so $H(1) = 9 - 8 - 1 = 0$. Another difference gives

$$\Delta H(n) = 8(9^n - 1).$$

Since $9 \equiv 1 \pmod 8$, the bracket is divisible by 8 for every n; hence $\Delta H(n)$ is divisible by 64. Telescoping and $H(1) = 0$ show $64 \mid H(n)$, and consequently $512 \mid \Delta F(n)$. Another telescoping step and $F(1) = 0$ prove the claim.

The two differences were chosen to peel away the polynomial terms and expose increasing powers of 2; no long binomial expansion of 9^n was required.

§1.4 Harmonic reflection modulo a prime

The rational-sum exercise asks us to write

$$S = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots - \frac{1}{1318} + \frac{1}{1319} = \frac{p}{q}$$

in lowest terms and prove $1979 \mid p$. The central observation is that

$$1979 - 1 = 1319 + 659.$$

Let $H_n = \sum_{j=1}^{n} 1/j$ with $H_0 = 0$. Separating odd and even denominators gives

$$S = H_{1319} - 2\sum_{j=1}^{659} \frac{1}{2j} = H_{1319} - H_{659}.$$

The integer 1979 is prime (trial division by primes at most $\sqrt{1979} < 45$ confirms this). Therefore every denominator from 1 to 1978 has an inverse modulo 1979; rational sums with these denominators make sense in the finite field $\mathbb{F}_{1979}$.

For any odd prime ℓ, pair j with $\ell - j$:

$$\frac{1}{\ell - j} \equiv -\frac{1}{j} \pmod{\ell}.$$

Summing over $1 \le j \le \ell - 1$ yields

$$H_{\ell-1} \equiv 0 \pmod{\ell}.$$

More precisely,

$$\begin{aligned} H_{\ell-1-m} &= H_{\ell-1} - \sum_{j=\ell-m}^{\ell-1} \frac{1}{j} \\ &\equiv -\sum_{j=1}^{m} \frac{1}{\ell - j} \equiv H_m \pmod{\ell}. \end{aligned}$$

Choose $\ell = 1979$, $m = 659$. Then $\ell - 1 - m = 1319$, so $H_{1319} \equiv H_{659} \pmod{1979}$ and hence

$$\boxed{S \equiv 0 \pmod{1979}.}$$

No denominator of S is divisible by 1979, and this property survives when the fraction is reduced to lowest terms. Consequently the reduced numerator p is divisible by 1979, as required.

The proof does not require constructing the astronomically large common denominator. The complementary harmonic intervals were deliberately chosen so that modular reflection pairs them.

§1.5 Prime-power valuations clarify the method

For a prime p, write $v_p(N)$ for the exponent of p in a nonzero integer N. Then

$$p^k \mid N \quad \Longleftrightarrow \quad v_p(N) \ge k.$$

The modulus $512 = 2^9$ is a valuation condition: after two finite-difference reductions, we extract three factors of 8, providing nine factors of 2. The harmonic problem instead uses a prime ℓ that is invertible in all denominators, so it is naturally solved by arithmetic in $\mathbb{F}_\ell$.

These two cases explain why modular arithmetic changes shape with the modulus. At a prime, inverses and field symmetries can be used; at a prime power, tracking the number of factors of the prime is often the right invariant.